Perfect—this is exactly how you build real Rust skill fast: **small, consistent challenges**.

Here's a **30-day Rust challenge plan** (1 per day or every 2 days). It's structured to move you from **beginner → job-ready fundamentals**.

🚀 **30-Day Rust Challenge Plan**

🟢 **Week 1 — Core Rust Foundations**

Focus: variables, control flow, ownership basics

Day 1: Hello CLI Tool

- Print user name from input
- Greet them dynamically

Day 2: Even or Odd Checker

- Input a number
- Determine if even or odd

Day 3: Temperature Converter

- Convert Celsius ↔ Fahrenheit

Day 4: Simple Calculator

- Support + - * /
- Use match

Day 5: Number Guessing Game

- Random number
- Loop until correct

Day 6: Multiplication Table Generator

- Input number
- Print table up to 10

Day 7: Word Counter

- Count words in a sentence

🟡 **Week 2 — Ownership, Structs, Enums**

Focus: Rust's core power

Day 8: String Reverser

- Reverse user input safely

Day 9: Palindrome Checker

- Check if a string reads same backward

Day 10: Struct Practice (User Profile)

- Create a User struct
- Print formatted data

Day 11: Rectangle Area Calculator

- Struct with width & height
- Add method area()

Day 12: Enum-Based Calculator

- Use enum for operations
- Cleaner than match-only logic

Day 13: Simple Banking System

- Struct: Account
- Deposit / withdraw
- Prevent overdraft

Day 14: To-Do List (Memory Only)

- Add / list tasks (no file yet)

🟠 **Week 3 — File I/O, Error Handling**

Focus: real-world Rust

Day 15: Save To-Do List to File

- Store tasks in a file

Day 16: Read File Content

- Read and print file contents

Day 17: Append to File

- Add new content without overwriting

Day 18: Basic Log Analyzer

- Count lines containing "ERROR"

Day 19: CSV Reader

- Parse simple CSV manually

Day 20: Error Handling Practice

- Use Result and match
- Handle invalid input safely

Day 21: Password Generator

- Use rand crate
- Generate secure passwords

🔵 **Week 4 — Intermediate Rust (Real Skills)**

Focus: collections, iterators, mini-systems

Day 22: Expense Tracker

- Track income/expenses
- Show total balance

Day 23: Frequency Counter

- Count how many times each word appears

Day 24: Command Parser

- Build mini CLI like:
- add task1
- list
- delete 1

Day 25: File Organizer

- Sort files into folders by type

Day 26: Simple HTTP Server

- Respond with "Hello World"

Day 27: Multi-Thread Counter

- Use threads to count numbers concurrently

Day 28: Chat App (Basic TCP)

- Send messages between two terminals

Day 29: Markdown → HTML Converter

- Convert headings (# Title) to <h1>

Day 30: Final Project (Mini System)

Choose one:

- Full CLI To-Do App (with file persistence)
- Expense Tracker CLI
- Notes App with search

💡 How to Use This Plan (Important)

Daily Routine (Best Practice)

- ⏱️ 1–2 hours max
- ✍️ Write code yourself (don't copy)
- 🧪 Break things intentionally
- 🔁 Refactor after it works

🧠 What You'll Gain After 30 Days

You'll understand:

- Ownership & borrowing (the hardest part 💀)
- Structs & enums (core Rust design)
- File handling (real-world skill)
- Error handling (Result, Option)
- CLI + system programming basics
- Intro to networking & concurrency

🔥 Want to go further?

I can:

- Turn this into a **trainer-ready curriculum (like your JS course)**
- Provide **starter templates for each day**
- Or create a **Rust exam + real-world project for your trainees**

Just say: *"Go deeper"* and I'll level this up 🚀